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MACROECONOMIC ACTIVITY AND INCOME DISTRIBUTION IN THE POSTWAR UNITED STATES

Alan S. Blinder and Howard Y. Esaki*

I. Introduction and Motivation

This paper is motivated by the frequent outcries against inflation on the grounds of its adverse effects on the distribution of income. Arthur Burns, for example, has asserted that "there can be little doubt that poor people . . . are the chief sufferers of inflation."[1] James Tobin (1972) has countered that "facile generalizations about the progressivity or equity of inflationary transfers are hazardous; certainly inflation does not merit the cliche that it is 'the cruelest tax'." He goes on: "Let us not forget that unemployment has distributional effects as well as deadweight losses."

This paper seeks to address this issue directly and quantitatively. How do inflation and unemployment affect the size distribution of income? While our principal concern is with cyclical influences, we felt it important to control for any trends that might be present in the income distribution data so as not to confuse cyclical and secular movements. This effort yielded some spillover benefits in shedding additional light on the trend in income inequality since World War II—a matter in some debate right now.[2]

While we are not the first to study the effects of the business cycle on the distribution of income,[3] our work differs from that of others in that (a) we do not impose any particular functional forms or measures of inequality on the income distribution data; (b) we obtain a disaggregated view of the effects of the cycle by income class; (c) we enquire into the effects of inflation as well as unemployment; (d) we base our analysis on the time series of shares of family income collected in the annual Current Population Survey (CPS).[4] While there are certain well-known problems with these data,[5] they are undoubtedly the best time series available for any long period of time. (The series begins in 1947.)

II. The Model, the Data, and the Estimation Techniques

The basic statistical model that we estimate is simply

$$S_i(t) = \alpha_i + \beta_i U(t) + \gamma_i \pi(t) + \delta_i T(t) + \epsilon_i(t), \qquad (1)$$

where $S_i(t)$ is the share of the ith quintile ($i = 1, \ldots, 5$) in the distribution of income among U.S. families in the tth year ($t = 1947, \ldots, 1974$); U is the overall unemployment rate; π is the rate of inflation as measured by the GNP deflator; and T is a linear time trend beginning with 1 in 1947. In addition, we estimated an equation of the same form for the share of the top 5% of families.[6]

Estimation was by ordinary least squares since more sophisticated techniques did not seem to be called for. First, there is no reason to expect any important reverse causation from the income distribution to unemployment or inflation.[7] Second, despite the fact that the left-hand variables are highly autocorrelated time series and that no lagged dependent variables were included as regressors, the residuals did not show evidence of autocorrelation. Third, heteroskedasticity would not normally be expected in a regression where none of the variables (save time itself) show much of a time trend. And, finally, ordinary least squares automatically imposes the cross-equation constraints:

$$\sum_1^5 \alpha_i = 1,$$

$$\sum_1^5 \beta_i = \sum_1^5 \gamma_i = \sum_1^5 \delta_i = 0,$$

$$\sum_1^5 \epsilon_i(t) = 0 \text{ for all } t.^8$$

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[1] From *Tax Review*, May 1968, as quoted in Palmer (1973).

[2] See the comprehensive survey by Taussig (1976).

[3] Other studies of the distributional impact of the business cycle are Beach (1973), Gramlich (1974), Metcalf (1972), Mirer (1973a, b) Schultz (1969) and Thurow (1970). Our work has most in common with that of Beach.

[4] All of the studies listed in footnote 3 share some of these characteristics, but do not share others.

[5] For example, property income is grossly underreported and the shifting nature of the family unit introduces an element of noncomparability over time.

[6] Regressions using some alternative measures of U and π were also run, with little change in the results.

[7] For example, Blinder (1975) found that the income distribution had almost no effect on aggregate consumption, and what effect it did have was in the perverse direction, i.e., consumption rose with income inequality. See also Metcalf (1972).

[8] The equations in (1) are actually a set of "seemingly unrelated regressions," but Zellner's (1962) technique reduces to ordinary least squares when the right-hand variables

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One quirk in the data merits discussion. In the early 1970s, the Bureau of the Census recomputed its historical series on quintile shares from the original microeconomic observations. However, it could not do this for years prior to 1958, and instead approximated the shares by fitting a density function to the *grouped* data.[9] So the pre-1958 data are much less accurate.

Several statistical tests were made to see whether the change in data construction techniques resulted in a spurious structural shift in equation (1). These tests showed that the coefficients of unemployment and inflation were stable over the entire period, while both the time trend and constant term shifted around 1958.[10] But this does not necessarily mean that data-construction methods were responsible for the shift. To study this further, we obtained some unpublished series in which the Census Bureau estimated quintile shares from *grouped* data for *every year* from 1947–1974 using a consistent technique. When the same tests of structural stability were performed on this consistent data, it again appeared that the constant and the coefficient of time shifted around 1958, while the coefficients of unemployment and inflation did not. Thus the shift may have been a genuine change in the economic system rather than in the data-processing methods.[11]

With some trepidation, then, we decided to use the more reliable data from 1958–1974 and introduce dummies into equation (1) to adjust for the structural shift, whatever its cause. Thus the basic model to be estimated is

$$S_i(t) = \alpha_i + \alpha'_i D(t) + \beta_i U(t) + \gamma_i \pi(t) + \delta_i T(t) + \delta'_i D(t)T(t) + \epsilon_i(t), \quad (2)$$

where $D(t)$ is a dummy variable equal to 0 in 1947–57 and 1 in 1958–74.[12]

III. The Basic Estimates

Table 1 presents the basic results obtained by estimating equation (2), and, for the fourth quintile only, also gives the results from estimating equation (1).[13] In

are identical in all equations. On this see Rao and Mitra (1971).

[9] The approximation uses a piecewise uniform distribution except at the upper tail, where a Pareto distribution is fit. For details see *Current Population Reports,* Series P-60, No. 97, pp. 169–170.

[10] For all five quintiles, the F-values for the test of stability of the unemployment and inflation coefficients were substantially less than the critical value for rejection of the hypothesis "no change in 1958" at the 10% confidence level. The hypothesis that both the constant and time trend were stable over the entire period could be rejected at the 1% confidence level for 4 of the 5 quintiles. Shifts in other years aside from 1958 were also tried, but 1958 seemed to give the best overall results. Details are available on request.

[11] The anti-poverty programs that began in the early 1960s are a possible explanation. It is worth noting that with the consistent data it was much less clear that the shift occurred in 1958 rather than in some other year.

[12] Regressions run using the consistent data for the entire period resulted in very minor changes in the sizes of the coefficients, and produced the same basic conclusions as equation (2).

[13] This is because the null hypothesis $\alpha'_4 = \delta'_4 = 0$ could not be rejected.

TABLE 1.—ESTIMATES FOR ALL FAMILIES: 1947–1974
(STANDARD ERRORS IN PARENTHESES)

Variable	(1.1) Lowest Fifth	(1.2) Second Fifth	(1.3) Middle Fifth	(1.4) Fourth Fifth	(1.5) Fourth Fifth	(1.6) Highest Fifth	(1.7) Top 5%
Constant	5.15[a]	12.36[a]	17.04[a]	23.23[a]	23.36[a]	42.19[a]	17.40[a]
	(.17)	(.20)	(.22)	(.22)	(.16)	(.48)	(.53)
Unemployment	−0.129[a]	−0.135[a]	−0.031	+0.042	+0.044	+0.272[a]	+0.053
	(.027)	(.030)	(.034)	(.034)	(.031)	(.074)	(.082)
Inflation	+0.031[b]	+0.010	−0.007	−0.023	−0.033[a]	−0.005	−0.008
	(.011)	(.013)	(.014)	(.014)	(.011)	(.031)	(.034)
Dummy	0.12	1.03[a]	1.18[a]	0.28	—	−2.65[a]	−1.61[b]
	(.20)	(.23)	(.26)	(.26)		(.55)	(.61)
Time	+0.014	+0.060[a]	+0.101[a]	+0.037[b]	+0.021[a]	−0.227[a]	−0.136[a]
	(.014)	(.015)	(.018)	(.018)	(.004)	(.038)	(.041)
Time × Dummy	+0.013	−0.088[a]	−0.118[a]	−0.024	—	+0.232[a]	+0.118[b]
	(.017)	(.019)	(.022)	(.022)		(.047)	(.052)
R^2	0.89	0.67	0.74	0.68	0.66	0.84	0.76
Standard Error	0.13	0.15	0.16	0.16	0.16	0.35	0.39
Mean of Left-hand Variable	5.07	12.18	17.62	23.75	23.75	41.39	16.12
Durbin-Watson	1.27	1.41	2.01	1.63	1.59	1.67	2.29

[a] Significant at 1% level by two-tailed t-test.
[b] Significant at 5% level by two-tailed t-test.

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general, the fits are rather good given that the lagged dependent variable does not enter the regression and that the equations omit every variable relevant to the income distribution except unemployment, inflation and time.

A very clear pattern of the incidence of unemployment by income class emerges. The lowest 40% of families loses most when U rises. Since the absolute losses are about the same (0.13 of a percentage point for each point increase in U), the lowest quintile loses relatively more of its share. What might loosely be called the "middle classes" (the 41st–80th percentiles) are almost unaffected, though the point estimates show the middle quintile losing a little to the fourth. The top of the income distribution naturally gains what the lower income classes lose. The coefficients show that the share of the highest quintile increases by the amount that the share of the two lowest quintiles decreases.

Compared to the sizeable unemployment coefficients, the coefficients for the inflation variable all look very close to (and most are insignificantly different from) zero. The only significant finding is that the income share of the poor increases during inflation, though the coefficient is only one-fourth the size of that of unemployment. For example, the sharp rise in the inflation rate between 1973 and 1974 (from 5.9% to 10%) added 0.13 percentage points to the share of the lowest quintile, according to the estimated coefficient. It appears that this small gain is made mainly at the expense of the fourth quintile; the inflation coefficient for this group just misses being significant in equation (1.4) and is significantly negative in equation (1.5).

The findings on the time trend are interesting in the context of the controversy that is raging over this issue. For three of the five quintiles, and for the top 5%, there is very strong evidence—in the form of highly significant coefficients on the interaction term $D(t)T(t)$—of a shift in the time trend in 1958. According to the estimates, between 1947 and 1957 the highest quintile, and especially the top 5%, suffered a steady and rather sizeable erosion in its income share while the 21st–80th percentiles, and especially the middle quintile, gained. After 1958, however, the pattern is completely different. There is almost no evidence of any trend in the upper 60% of the income distribution.[14] The minor trend redistribution that did go on was a transfer from the second quintile (the working poor?) to the lowest quintile (the poor). A speculative explanation might focus on the rising payroll tax and burgeoning income-support system.

Since the accuracy of the CPS data on income shares is so questionable for the period prior to 1958, table 2 presents regression estimates using only the "good data" from 1958 to 1974. The coefficients of unemployment show a slight strengthening of the picture that emerged from table 1: the lowest 60% (and especially the lowest 40%) lose ground to the highest 40% (and especially the highest 20%) when the unemployment rate rises. The signs of the coefficients are always the same as those obtained for the longer sample, and the magnitudes are a bit larger. The standard errors seem hardly to have suffered from the drop in the number of degrees of freedom from 22 to 13.

The inflation coefficients, however, do look somewhat different. While most of them fail to attain significance at conventional levels, the suggestion is that inflation in 1958–74 increased the share of the 21st–60th percentiles at the expense of the upper 20%, and espe-

[14] An exception is some weak evidence of a slow erosion in the share of the middle quintile.

TABLE 2.—ESTIMATES FOR ALL FAMILIES: 1958–1974
(STANDARD ERROR IN PARENTHESES)

Variable	Lowest Fifth	Second Fifth	Middle Fifth	Fourth Fifth	Highest Fifth	Top 5%
Constant	5.28[a]	13.74[a]	18.68[a]	23.57[a]	38.65[a]	14.85[a]
	(.31)	(.30)	(.25)	(.28)	(.76)	(.64)
Unemployment	−0.146[a]	−0.154[a]	−0.071[b]	+0.052	+0.334[a]	+0.136[c]
	(.035)	(.035)	(.029)	(.031)	(.087)	(.073)
Inflation	+0.009	+0.064[b]	+0.038	+0.006	−0.122	−0.103
	(.029)	(.028)	(.023)	(.026)	(.070)	(.060)
Time	+0.034[b]	−0.050[a]	−0.038[a]	+0.002	+0.054	+0.025
	(.013)	(.013)	(.011)	(.012)	(.033)	(.028)
R^2	0.85	0.68	0.61	0.25	0.58	0.41
Standard Error	0.13	0.13	0.11	0.11	0.32	0.27
Mean of Left-hand Variable	5.25	12.17	17.70	23.89	41.02	15.69
Durbin-Watson	1.38	1.15	1.33	1.51	1.33	2.05

[a] Denotes significant at 1% level by two-tailed t-test.
[b] Denotes significant at 5% level by two-tailed t-test.
[c] Denotes significant at 10% level by two-tailed t-test.

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cially the top 5%. While this is a substantially different pattern from that of table 1, both agree that inflation is equalizing rather than disequalizing. Unfortunately, the smaller sample size has blown up the standard errors enough so that not much can be said with confidence.

The time trends estimated for the 1958–1974 period in table 1 also change somewhat when the regression is limited to these years. In table 1 it appears that the second quintile was yielding ground to the first during 1958–74. But things look rather different in table 2, where the 21st–60th percentiles seem to be losing ground to the two tails.[15]

IV. The Shifting Age Distribution and the Trend in Inequality

A number of authors, including Kuznets (1974) and Paglin (1975), have pointed to the importance of changes in the U.S. age distribution for the trend in income inequality. Indeed, Paglin has gone so far as to maintain that demographic change—in particular, the increasing numbers of young and old household heads—has hidden a rather marked downward trend in inequality. For example, the youngest (14–24 years) and oldest (65+ years) age groups, who always have the lowest incomes, comprised only 16.6% of family heads in 1947 but 21.9% in 1974.

If it is true that income distributions for narrowly-defined age groups have been equalizing, while the increasing weight given to very young and very old households has counteracted this and given a false impression of stability in the overall income distribution, then analysis of an income distribution *corrected for demographic change* might show a dramatically different time trend than our regressions in table 1. It is even possible that making this correction could alter our conclusions about the cyclical sensitivity of the income distribution. For both of these reasons, then, it seems important to at least try to control for demographic factors.

Shifts in the age distribution alter the distribution of *annual* income even if there is no change in the distribution of *lifetime* income because the typical age-income profile is concave, and because within-cohort inequality differs by cohort. By creating a hypothetical income distribution series based on *fixed population proportions,* we abstract from both of these phenomena, while leaving in the data any changes in either the age-income profile or within-cohort inequality that actually occurred. Specifically, our age-corrected income distributions are constructed from official data as follows. The CPS reports annual distributions for each age group, $f_t^i(y)$, which are combined into an overall distribution:

$$f_t(y) = \sum_{i=1}^{6} w_i(t) f_t^i(y),$$

where $w_i(t)$ is the fraction of families in the ith age group in year t. The age-corrected income distribution would be

$$\bar{f}_t(y) = \sum_{i=1}^{6} w_i f_t^i(y),$$

where the w_i are the population weights of some base year.

While the $\bar{f}_t(y)$ are easy to construct, difficulties arise in measuring shares since the data give only scattered points on the cumulative distribution functions (from 13 to 18 points, depending on the year), and since age-specific mean incomes by income class are unknown (except for recent years). Two alternative approximation procedures were adopted and are described in the appendix. One amounts to approximating the distribution as a step function with a Pareto upper tail; the other amounts to approximating the entire distribution as the sum of several lognormal distributions.

The data generated by both of these approximation procedures were used to re-estimate equation (2). Neither method of age correction led to any major changes in the coefficients shown in table 1. In the case of unemployment and inflation, this came as no surprise. However, in the case of the time trend, it was rather surprising, since demographic change is often cited as a cause of increasing income inequality. We conclude—rather tentatively, given the rough nature of our corrections for the age distribution—that the postwar demographic shifts have had much less effect on the trend in inequality than is currently believed.

V. Summary of Conclusions

Of all our findings, the one unequivocal message seems to be that the incidence of unemployment is quite regressive. We estimate that each one percentage point rise in the unemployment rate takes about 0.26%–0.30% of the national income away from the lowest 40% of the income distribution and gives it to the richest 20%. This estimate is quite robust to a number of minor changes in the specification.

[15] The regressions of tables 1 and 2 were also run with families disaggregated by race. The results for white families were almost identical with those for all families—hardly surprising since white families comprised 89% of the total in 1974. For black families, the story was slightly different. The coefficients of unemployment tell the same broad *qualitative* story as they do for whites, but show a much greater cyclical sensitivity, and differ in details of incidence. And for blacks, unlike for whites, inflation does appear to be a rather regressive tax. Although the lowest quintile of black families neither gains nor loses from inflation, the 21st–60th percentiles lose ground to the 81st–95th percentiles. Complete results are available from the authors upon request.

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While the findings on inflation are less firm, it appears that inflation is a slightly progressive tax in that the poor and middle classes lose relatively less than the rich. More to the point, the effects of inflation on the income distribution simply are much less important than those of unemployment.

We find evidence of a significant change in the trend in inequality around 1958. While this finding is troublesome because that year marks a change in the Bureau of the Census' methods for computing the quintile shares, we find a similar shift in a consistent data set where there is no methodological change. Before 1958, it appears, the middle 60% of families were gaining at the expense of the richest 20%. Since 1958, however, the poorest 20% of families have been gaining at the expense of the next poorest 20% (and perhaps also the middle 20%). The top 20% of families have ceased losing ground, and may even be increasing their share.

Surprisingly, none of this is changed much by adjusting the income shares for changes in the age distribution—by either of two approximation procedures.

APPENDIX

Description of Procedures Used to Calculate Age-corrected Income Distributions

In the first procedure we calculated quintile shares in a manner very close to that used by the Census Bureau. For all but three of the income classes, the density function was approximated by a piecewise uniform density. For the top open-ended interval, we adhered to the Census method of fitting a Pareto tail to estimate the mean income in the class. (See U.S. Bureau of the Census (1967), pp. 33–35.) For the highest closed interval, a linear density with a negative slope such that the right number of people were assigned to the interval was used. A similar procedure was followed for the bottom interval.

The second procedure amounted to assuming that the income distribution *within* each age group was lognormal. To apply this method we first estimated the parameters of the two-parameter lognormal distribution for each of the six age groups for each of the 28 years. This was done by a series of regressions that exploited the fact that[1]

$$\log y(q) = \mu + \sigma z(q), \tag{3}$$

where

μ,σ = the two parameters to be estimated
$y(q)$ = the income level at the qth percentile of the distribution
$z(q)$ = the qth percentile of a unit normal deviate, i.e., $Pr(x \leq z(q)) = q$ if x is a unit normal deviate.

Depending on the number of income brackets, there were anywhere from 12 to 17 observations available through which to fit equation (3). In order that the regression not give undue weight to the two tails (where the lognormal does not fit well), the equations were estimated by weighted least squares. The weights were (roughly) the height of the density in each income range. Once all the means and log variances had been estimated, a program written by Sherman Robinson (1976) was used to aggregate the six age-specific distributions for a given year into one overall distribution. Since this overall distribution is not itself lognormal, quintile shares were calculated numerically by the program.

Neither of these approximations fits the data exactly and, in fact, both introduce large potential errors in the tails. Realizing this, we decided to apply the approximation techniques to *both* the actual income distributions, f_t, *and* the age-corrected distributions, $\bar{f}_t$. This gave us two sets of income shares for each approximation method: $s_i(t)$ from the actual data, and $\bar{s}_i(t)$ from the hypothetical data. The *differences*, $\bar{s}_i(t) - s_i(t)$, were taken to be a pure correction for the age distribution. So the age-corrected shares $S'_i(t)$ were defined as

$$S'_i(t) = S_i(t) + (\bar{s}_i(t) - s_i(t))$$

where $S_i(t)$ are the official data. In general, the age corrections by either method were quite small. Since 1960 was chosen as the base year, the corrections amount to putting more old and young families into the pre-1960 distributions and taking some out of the post-1960 distributions. Thus the correction typically equalizes the distribution in recent years and disequalizes it in early years.[2]

[1] See Aitchison and Brown (1957), p. 31.

[2] Details of the computations, including the age corrections themselves and the regressions run with the $S'_i(t)$, are available on request.

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INFLATION AND RELATIVE PRICE DISTORTIONS: THE CASE OF HOUSING

J. R. Kearl*

I. Introduction

It is often suggested that perfectly anticipated inflation will cause no distortions of the real side of an economy except as such inflation redistributes income. This is interpreted to mean that relative prices are independent of changes in the price level that are anticipated correctly. Unanticipated price level changes may change relative prices as decision makers are in a sense "fooled" and make decisions based on inappropriate anticipations. Whether or not a policy maker can exploit these is central to the now long running debate on the meaning (and existence) of a Phillips Curve in the labor market.

While the investigation and debate over these issues in the labor market has been, to say the least, extensive, economists have shown less concern over the impact of inflation elsewhere in the economy. It is alleged, for example, that pure inflation-induced changes in nominal interest rates should not affect housing demand. Thus, "if market rates rise and are expected to remain permanently at their new level as a result of an increase in the anticipated rate of long-run inflation, there is no reason to believe that the demand for housing is permanently reduced . . . it is only at higher real rates (that) fewer houses are built" (Arcelus and Meltzer, 1973).

Consider, however, the following example. A household contemplates a 30-year, $30,000 mortgage to finance a dwelling unit. With no inflation and a real rate of say 3%, annual payments necessary for full amortization would amount to $1,517. If a 2% inflation were anticipated, the contract annual payment would jump to $1,931, nearly a 25% increase. A 6% inflation would result in a 9% contract rate and payment of $2,895, nearly double the no inflation payment.

The expectation of inflation does not cause the household's real financial position to deteriorate, yet it increases the real burden of debt service in the present since it is only in the *future* that inflation may deliver higher nominal incomes with which to make the *current* (and future) inflation-induced higher payments.[1]

Tucker (1974) notes that from 1965 to 1973 wage and salary income increased 57% (12% in real terms) while initial monthly payments, despite lengthening of amortization periods, increased 85%. The fraction of income taken up by monthly payments would have been 16% higher for the average household in 1973 over a similar purchase in 1965, this, despite real incomes increasing by 12%.[2]

Poole (1972) first suggested that the phenomenon illustrated by these examples was likely to cause distortions in a market that utilizes the long-term, level nominal payment, fully amortized mortgage for housing acquisition and would thus possibly interfere with efficient allocation of savings and with capital accumulation. Specifically, I postulate that such distortions are in the nature of relative price changes.

The significance of this phenomenon is an empirical question and one of some importance considering the several efforts being made to allow for mortgage innovation and the many efforts by the government to

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[1] Should a contract be offered that allowed one to borrow against future inflated income (an indexed or price-level adjusted contract), then this problem would be eliminated. But this is precisely the point, such contracts are not being offered.

[2] This example assumes the same percentage down payment was made in each year and that housing prices were not going up faster than the inflation rate, which they were not.

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